

An extended generalized Pareto regression model for count data

Touqeer Ahmad¹, Carlo Gaetan² and Philippe Naveau³
(touqeer.ahmad@ensai.fr)

¹ CREST, ENSAI, France. ² Università Ca' Foscari, Venezia, Italy

³ LSCE, Paris, France.

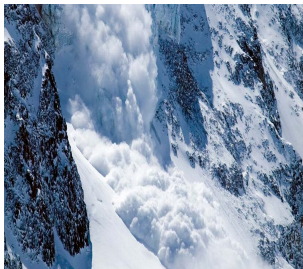
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- ② Introduction & background
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Motivation: extremes of avalanches

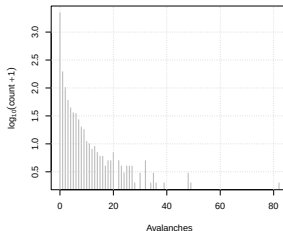
- **Snow avalanches** are among the most destructive natural hazards in mountainous regions and threatening to:
 - human lives
 - ecosystems
 - built structures and
 - landscapes



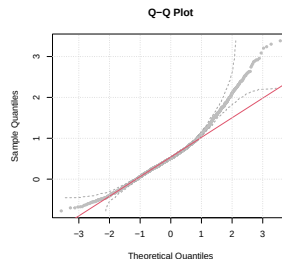
<https://www.sudouest.fr/tourisme/pyrenees/meteo-france-alerte-sur-les-avalanches-3531479.php>.

Motivation: fit of existing model

- Zero-inflated negative binomial regression model fitted to avalanche counts of Haute-Maurienne massif.



(a)



(b)

Motivation: encounter the problem

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- We required a model that correctly captures the lower and upper tail behavior.
- An extreme value model, or a heavy-tailed distribution, may be helpful for such extreme events modeling.

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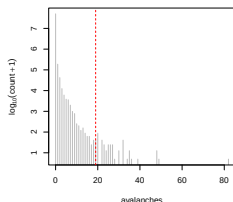
Role of Extreme Value Theory

- On possibility is to model threshold exceedances using **discrete generalized Pareto distribution** (DGPD) Hitz et al. (2017).

The DGPD is defined by discretizing

$$\mathcal{F}(k) = 1 - (1 + k\xi/\sigma)^{-1/\xi}$$

$$\Pr(Y-u = k | Y \geq u) = \mathcal{F}(k+1; \sigma, \xi) - \mathcal{F}(k; \sigma, \xi), \quad k \in \mathbb{N}_0 \quad (1)$$



Problems with DGPD fitting

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- Optimal threshold selection is problematic.
- Losing a large amount of data when using high quantile for threshold selection.
- We required a **flexible model** that accurately models count data across different scenarios:
 - ① the entire data with a heavy tail
 - ② the data with excessive zeros and heavier tail together
 - ③ the data above a very low threshold (Ahmad and Arshad, 2024)

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Discrete extended generalized Pareto distribution (DEGDP)

- Construction is based on the integral transform idea to simulate GPD random draws Naveau et al. (2016), that is

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where G is a CDF on $[0, 1]$ and $U \sim \mathcal{U}(0, 1)$.

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where G is a CDF on $[0, 1]$ and $U \sim \mathcal{U}(0, 1)$.

- The **key problem** is to find a class for G which preserves the upper tail behavior with shape parameter ξ and also controls the lower tail behavior

DEGPD

- Naveau et al. (2016) defined restrictions for validity of G families. For instance, the tail of G denoted by $\bar{G} = 1 - G$ has to satisfy

$$\lim_{u \rightarrow 0} \frac{\bar{G}(1-u)}{u} = a, \text{ for some finite } a > 0 \text{ (upper tail behavior),}$$
$$\lim_{u \rightarrow 0} \frac{G(u)}{u^\kappa} = c, \text{ for some finite } c > 0 \text{ (lower tail behavior).} \quad (3)$$

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- We start by considering the CDF $G\{\mathcal{F}(\cdot; \sigma, \xi)\}$ of EGPD where G meets the conditions (3). To model non-negative integer data, we discretize the CDF by

$$\Pr(Y = k) = G\{\mathcal{F}(k+1; \sigma, \xi)\} - G\{\mathcal{F}(k; \sigma, \xi)\}, \quad k \in \mathbb{N}_0 \quad (4)$$

The distribution will be called **DEGPD**.

Zero-inflated DEGPD

Zero inflation problem along with heavier tail...

Let Y be observed with an **excessive number of zeros** relative to those observed under the DEGPD. The zero-inflated distribution (**ZIDEGPD**) is defined in a straightforward way as:

$$\Pr(Y = m) = \begin{cases} \pi + (1 - \pi)G\{\mathcal{F}(1, \sigma, \xi)\} & m = 0 \\ (1 - \pi)[G\{\mathcal{F}(m + 1, \sigma, \xi)\} - G\{\mathcal{F}(m, \sigma, \xi)\}] & m \in \mathbb{N} \end{cases} \quad (5)$$

where $0 \leq \pi \leq 1$ and the remaining parameters would be the same as DEGPD.

It turns out that the CDF of ZIDEGPD is

$$\Pr(Y \leq m) = \pi + (1 - \pi)G\{\mathcal{F}(m + 1, \sigma, \xi)\}, \quad m \in \mathbb{N}_0 \quad (6)$$

G families

- For G , we use four parametric expressions $G(\cdot, \psi)$, already proposed in Naveau et al. (2016), which preserve the upper tail behavior with shape parameter ξ .

- $G(u; \psi) = u^\kappa, \quad \psi = \kappa > 0$

- $G(u; \psi) = 1 - D_\delta \left\{ (1 - u)^\delta \right\} \quad \psi = \delta > 0$

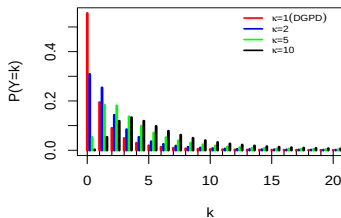
where D_δ is the CDF of beta distribution with parameters $1/\delta$ and 2.

- $G(u; \psi) = \left[1 - D_\delta \left\{ (1 - u)^\delta \right\} \right]^{\kappa/2} \quad \psi = (\delta, \kappa) > 0$

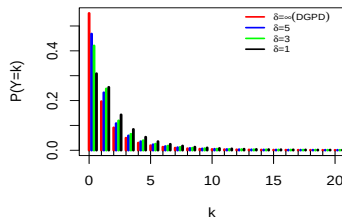
- $G(u; \psi) = pv^{\kappa_1} + (1 - p)v^{\kappa_2} \quad \psi = (p, \kappa_1, \kappa_2), \kappa_1, \kappa_2 > 0 \text{ and } p \in [0, 1], \kappa_1 \leq \kappa_2$

PMF behaviors of DEGPD

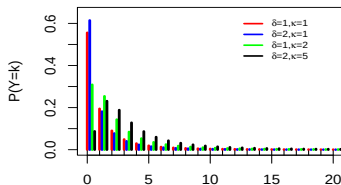
Model type (i) $G(u; \psi) = u^\kappa$



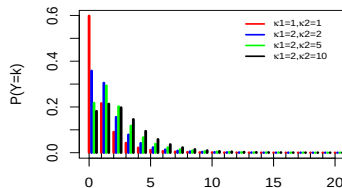
Model type (ii) $G(u; \psi) = 1 - D_\delta((1-u)^\delta)$



Model type (iii) $G(u; \psi) = (1 - D_\delta((1-u)^\delta))^{\kappa/2}$



Model type (vi) $G(u; \psi) = pu^{\kappa_1} + (1-p)u^{\kappa_2}$



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- In the sequel, we denote the vector of parameters $(\pi, \xi, \sigma, \psi^T)^T$ with $\theta = (\theta_1, \dots, \theta_d)^T$
- The parameters of the distribution of Y may depend on some covariates $\mathbf{x}$, i.e. $\theta(\mathbf{x}) = (\theta_1(\mathbf{x}), \dots, \theta_d(\mathbf{x}))^T$.

Generalized additive modeling

- For relating the distributional parameters $(\theta_1(\mathbf{x}), \dots, \theta_d(\mathbf{x}))$ to the covariates, we consider additive predictors of the form

$$\eta_i(\mathbf{x}) = s_{i1}(\mathbf{x}) + \dots + s_{iJ_i}(\mathbf{x}) \quad (7)$$

where $s_{i1}(\cdot), \dots, s_{iJ_i}(\cdot)$ are smooth functions of the covariates $\mathbf{x}$.

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- The predictors are linked to the distributional parameters via known monotonic and twice differentiable link functions $h_i(\cdot)$.

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- In the case of model with $G(u; \psi) = u^\kappa$, common link functions are

$$\begin{aligned} \xi(\mathbf{x}) &= \exp(\eta_\xi(\mathbf{x})), & \sigma(\mathbf{x}) &= \exp(\eta_\sigma(\mathbf{x})), & \kappa(\mathbf{x}) &= \exp(\eta_\kappa(\mathbf{x})), \\ \pi(\mathbf{x}) &= \exp\left(\frac{\eta_\pi(\mathbf{x})}{1 + \eta_\pi(\mathbf{x})}\right) \end{aligned}$$

Generalized additive modeling

- The functions s_{ij} in (7) are approximated in terms of basis function expansions,

$$s_{ij}(\mathbf{x}) = \sum_{k=1}^{K_{ij}} \beta_{ij,k} B_k(\mathbf{x}), \quad (9)$$

where $B_k(\mathbf{x})$ are the basis functions and $\beta_{ij,k}$ denote the corresponding basis coefficients.

- The basis function expansions can be written as $s_{ij}(\mathbf{x}) = \mathbf{t}_{ij}(\mathbf{x})^T \boldsymbol{\beta}_{ij}$ where $\mathbf{t}_{ij}(\mathbf{x})$ is still a vector of transformed covariates that depends on the basis functions and $\boldsymbol{\beta}_{ij} = (\beta_{ij,1}, \dots, \beta_{ij,K_{ij}})^T$ is a parameter vector to be estimated.

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Estimation

- The likelihood function for ZIDEGPD (5) is written as.

$$\begin{aligned}
 l(\beta) = & \sum_{i=1}^n l_0(y_i) \log [\pi + (1 - \pi)G(\mathcal{F}(1; \sigma, \xi); \kappa)] \\
 & + \sum_{i=1}^n (1 - l_0(y_i)) \log(1 - \pi) \times \left[G(\mathcal{F}(y_i + 1; \sigma, \xi); \kappa) \right. \\
 & \left. - G(\mathcal{F}(y_i; \sigma, \xi); \kappa) \right], \tag{10}
 \end{aligned}$$

where β collects all unknown coefficient β_{ijk} of the basis expansions.

- The penalized log-likelihood function for the latter models reads:

$$l_p(\beta) = l(\beta) - \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^{J_i} \beta_{ij}^T \mathbf{G}_{ij}(\lambda_{ij}) \beta_{ij} \quad (11)$$

where $l(\beta)$ is the log-likelihood function (10).

- To fit ZIDEGPD with GAM forms, we have written an R code that implements the distributions as “new families” for evgam R package (Youngman, 2022).¹

¹An example of R code with the name “Fit_degpd_zidegpd.R” and the complete source code of the function is provided on the GitHub <https://github.com/touqeerahmadunipd/degpd-and-zidegpd>

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Application to avalanches data with environmental covariates

- We consider in particular **three-day observation window** of the daily number of avalanche events recorded from February 1982 to April 2021 (October 15th and May 15th). Further, we consider the following covariates

Name	Definition
WS	maximum wind speed at 10 meters (m/s)
PREC	precipitation (mm/day)
MxT	maximum temperature at 2 Meters (C°)
MnT	minimum temperature at 2 Meters (C°)
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- Three-day summaries are calculated by averaging RH and PREC and taking the maximum for MxT and WS and the minimum for MnT.

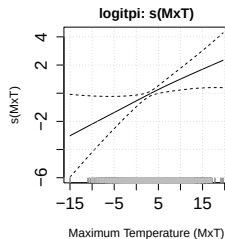
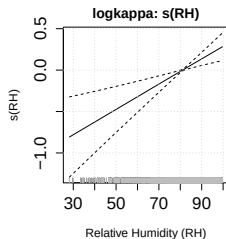
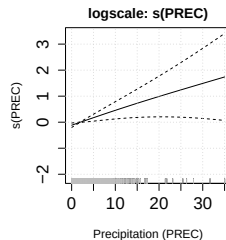
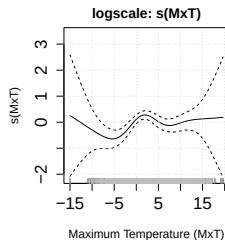
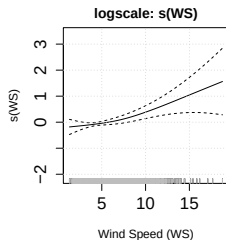
Selection of covariates

- After comparing different combinations of the covariates, we found WS, MxT, PREC, and RH are more appropriate to use as covariates.
- For instance, it turned out for the model $G(u, \Psi) = u^\kappa$ with the lowest AIC are

$$\begin{aligned} \text{Model type (1)} : \quad \kappa &= s(RH); \quad \sigma = s(WS) + s(MxT) \\ &+ s(PREC); \quad \xi = cst \\ \text{logit}(\pi) &= s(MxT) \end{aligned}$$

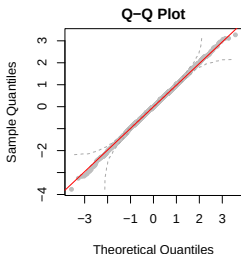
where $s(\cdot)$ indicates the smoothed predictor.

Non-parametric effects of covariates

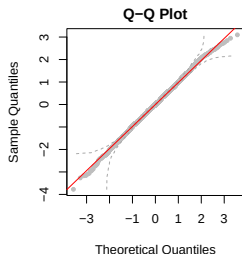


Model adequacy

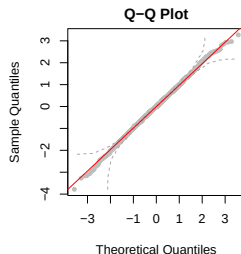
- To assess the proposed model and compare it with the existing models, we use randomized residuals analysis by following (Dunn and Smyth, 1996)



Model (i)



Model (ii)



Model (iii)

Goodness-of-fit

Table 1: AIC values and Anderson-Darling normality test statistics of the different ZIDEGPD models.

Model	AIC	Statistics	P-value
(i)	6121	0.287	0.623
(ii)	6151	0.367	0.432
(iii)	6152	0.315	0.543

Importance and further applications

- Motivated by the avalanche real data example, we attempt to properly model the lower limit of the distribution and the upper tail without neglecting the bulk part of the data.

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- Motivated by the avalanche real data example, we attempt to properly model the lower limit of the distribution and the upper tail without neglecting the bulk part of the data.
- Non-additive models can be fitted using thin-plate splines for isotropic spatial dependence or tensor product splines for covariate interactions.
- One can propose a new model if the new G-family fulfills the condition (3).

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Thank you for your attention!